

Introduction

Predicting customer churn is vital in helping financial institutions maintain long term relationships with clients and reduce revenue loss. In this project, the objective was to build a robust machine learning model capable of identifying customers who are likely to discontinue their relationship with Lloyds Banking Group. The report shows the process of selecting a suitable machine learning algorithm, preparing and balancing the dataset, training the models and evaluating their performance, and providing insights that can be applied within the business context.

To achieve this goal, three machine learning algorithms were used which are the Random Forest, Support Vector Machine and XGBoost. These models were chosen because they each have their own unique advantages that address different characteristics of the dataset. Random Forest and XGBoost were chosen as they are both ensemble algorithms based on decision trees, which makes them a good choice for structured data and capable of capturing nonlinear relationships. On the other hand, SVM serves as a comparative machine learning model that is not tree-based but excels in handling high-dimensional features and separating classes even when features are not correlated to one another. Testing these three algorithms allowed for a comprehensive assessment of their suitability for a churn prediction task, especially as the dataset showed significant imbalance between churn and non-churn customers.

Data Preprocessing

Even though the data was cleaned and we made new features from the task 1, for this project, we used the imbalance-learn python library in order to use SMOTE or the Synthetic Minority Over-sampling Technique which is a method used in machine learning to balance imbalanced datasets by creating new, synthetic samples of the minority class and we also used the Random Under Sample method of imbalance-learn that is used to address class imbalance by randomly selecting and deleting examples from the majority class until a more balanced class distribution is achieved.

Before building the models, it was important to address the issue of class imbalance, which can heavily distort the predictions of the machine learning algorithms. Although the dataset had already gone to data cleaning and feature engineering in Task 1, additional steps were required to ensure that the minority class which are the churned customers was adequately represented during training.

To achieve this, the Synthetic Minority Oversampling Technique or SMOTE was used. SMOTE works by creating synthetic examples of the minority class through interpolation, enabling the model to learn the characteristics of churned customers without simply duplicating minority samples. After applying SMOTE, Random Under-Sampling was used to reduce a portion of the majority class. This two-step process created a more balanced dataset that reduced the likelihood of the models becoming biased toward predicting non-churn outcomes. Once the

dataset was balanced, it was split into training and testing sets using stratified sampling to preserve the original distribution of the target class.

Model Building

Model	Churn Precision	Churn Recall	F1-Score	ROC-AUC	Notes
Random Forest	0.19	0.68	0.30	0.35	Best recall but low precision
SVM	0.22	0.56	0.32	0.46	Most balanced overall
XGBoost	0.16	0.37	0.23	0.44	Underperformed in recall

Three machine learning algorithms were trained on the processed dataset which are Random Forest, Support Vector Machine and XGBoost. Each model was selected based on its strengths and relevance to the churn prediction task.

The Random Forest classifier was trained using class weights set to “balanced”, which ensures that the algorithm penalizes misclassification of the minority class more heavily. This encouraged the model to focus more on identifying churners. Meanwhile, the Support Vector Machine model was trained with probability estimates enabled and with balanced class weights as well, allowing it to better separate the two classes through its margin-based decision boundary. Lastly, the XGBoost is a powerful gradient boosting method known for strong performance on tabular data, and was tuned with parameters like adjusted learning rate, increased depth and scale_pos_weight to compensate for the class imbalance.

To ensure fairness, each model was trained on the same dataset and assessed against the same testing set. The purpose of evaluating the three algorithms was to identify which one had the highest predicted performance and most closely matched the needs of a commercial environment.

Model Evaluation

After training the models, each was evaluated using metrics such as precision, recall, F1-score and ROC-AUC. These metrics provides understanding of how the models performed, especially when dealing with imbalanced data.

The Random Forest model showed strong recall for the minority class, detecting most of the customers who were likely to churn. However, this came with low precision, meaning that although the model correctly identified many churners, it also incorrectly classified many

non-churners as churners. This produces a large number of false positives. While this may burden the company with unneeded follow-up actions, it does ensure that fewer churners go missed.

The SVM classifier outperformed the other two models in terms of balance. It had higher accuracy, a stronger F1 score, and the greatest ROC-AUC score. Despite having a lower recall rate than Random Forest, it detected more than half of the churners in the sample. Overall, the SVM achieved the ideal balance of identifying churners while avoiding unnecessary false alarms.

XGBoost, despite being a powerful algorithm, underperformed the other two models in this particular example. Its memory and F1-score were significantly lower, indicating that it struggled to distinguish between churners and non-churners successfully. XGBoost may improve with more tuning, however based on the current findings, it is not the best-performing model.

Across all evaluations, the Support Vector Machine is the most effective model for this dataset. Its ability to separate classes more reliably and provide a balanced performance across all metrics made it the strongest candidate for deployment.

Conclusion

After comparing the performance of Random Forest, XGBoost, and Support Vector Machine (SVM) on the customer churn dataset, SVM was chosen as the primary algorithm for the finished model. Although Random Forest and XGBoost were both good, especially given their strengths as ensemble tree-based classifiers, SVM performed more balanced and consistently across the study's evaluation metrics. SVM's ability to create appropriate decision boundaries even when the dataset contains non-linearly related features was very useful for this churn prediction challenge. Its usage of kernel functions enabled the model to better adapt to nonlinear patterns in the data while avoiding overfitting than the other approaches. In addition, despite the data imbalance, SVM demonstrated good generalization and consistency, giving it a reliable alternative for real-world prediction scenarios. For these reasons, SVM was determined as the most appropriate algorithm and chosen as the principal model for the project's deliverables.

Business Application

Choosing SVM as the final model for the churn prediction system brings meaningful and practical value to business decision-making processes. With its capability to create precise and robust classification boundaries, SVM enables the organization to accurately distinguish customers who are likely to churn from those who are not, even when the differences between the two groups are subtle. This level of predictive dependability enables businesses to identify at-risk consumers earlier, which leads to more proactive and cost-effective retention initiatives. Because SVM works efficiently with high-dimensional data and eliminates the effects of noise, it provides businesses with a reliable and efficient model for operational deployment.

Furthermore, the clarity of SVM's decision-making process—especially when combined with feature analysis techniques helps companies determine which customer traits have the most influence on churn behavior. This enables corporate teams to create more focused marketing campaigns, improve service quality, tailor engagement activities, and deploy resources more strategically. Finally, the use of SVM offers the company with a consistent and effective prediction tool that promotes long-term customer retention and improves overall customer relations.